WEEK - 03

a) Develop a Java program to create an abstract class EnergySource with members Source ID, Source Name, and Energy Generated (in kWh). Include an abstract method calculateEfficiency(). Derive the classes SolarEnergy and WindEnergy from EnergySource and implement the method based on the following criteria: For SolarEnergy, efficiency is calculated as (Energy Generated / 5000) × 100. For WindEnergy, efficiency is calculated as (Energy Generated / 8000) × 100. Display source details and efficiency percentage. Demonstrate dynamic method dispatch using an EnergySource reference.

```
abstract class EnergySource {
    int sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(int id, String name, double energy) {
        sourceId = id;
        sourceName = name;
        energyGenerated = energy;
    }

    abstract double calculateEfficiency();

    void display() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " + energyGenerated + " kWh");
        System.out.println("Efficiency: " + calculateEfficiency() + "%");
        System.out.println();
    }
```

```
}

class SolarEnergy extends EnergySource {

    SolarEnergy(int id, String name, double energy) {
        super(id, name, energy);
    }

    double calculateEfficiency() {
        return (energyGenerated / 5000) * 100;
    }
}

class WindEnergy extends EnergySource {

    WindEnergy(int id, String name, double energy) {
        super(id, name, energy);
    }

    double calculateEfficiency() {
        return (energyGenerated / 8000) * 100;
    }
}

public class Main {
    public static void main(String[] args) {

        EnergySource e;
```

```
        e = new SolarEnergy(101, "Solar Panel", 4000);
        System.out.println("Solar Energy Details:");
        e.display();

        e = new WindEnergy(102, "Wind Turbine", 6000);
        System.out.println("Wind Energy Details:");
        e.display();

        System.out.println("NAME: Deekshitha H S");
        System.out.println("USN: 1BM26CS605-T");
    }
}
```

OUTPUT

```
 ----jGRASP exec: java Main
Solar Energy Details:
Source ID: 101
Source Name: Solar Panel
Energy Generated: 4000.0 kWh
Efficiency: 80.0%

Wind Energy Details:
Source ID: 102
Source Name: Wind Turbine
Energy Generated: 6000.0 kWh
Efficiency: 75.0%

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 ----jGRASP: Operation complete.
```